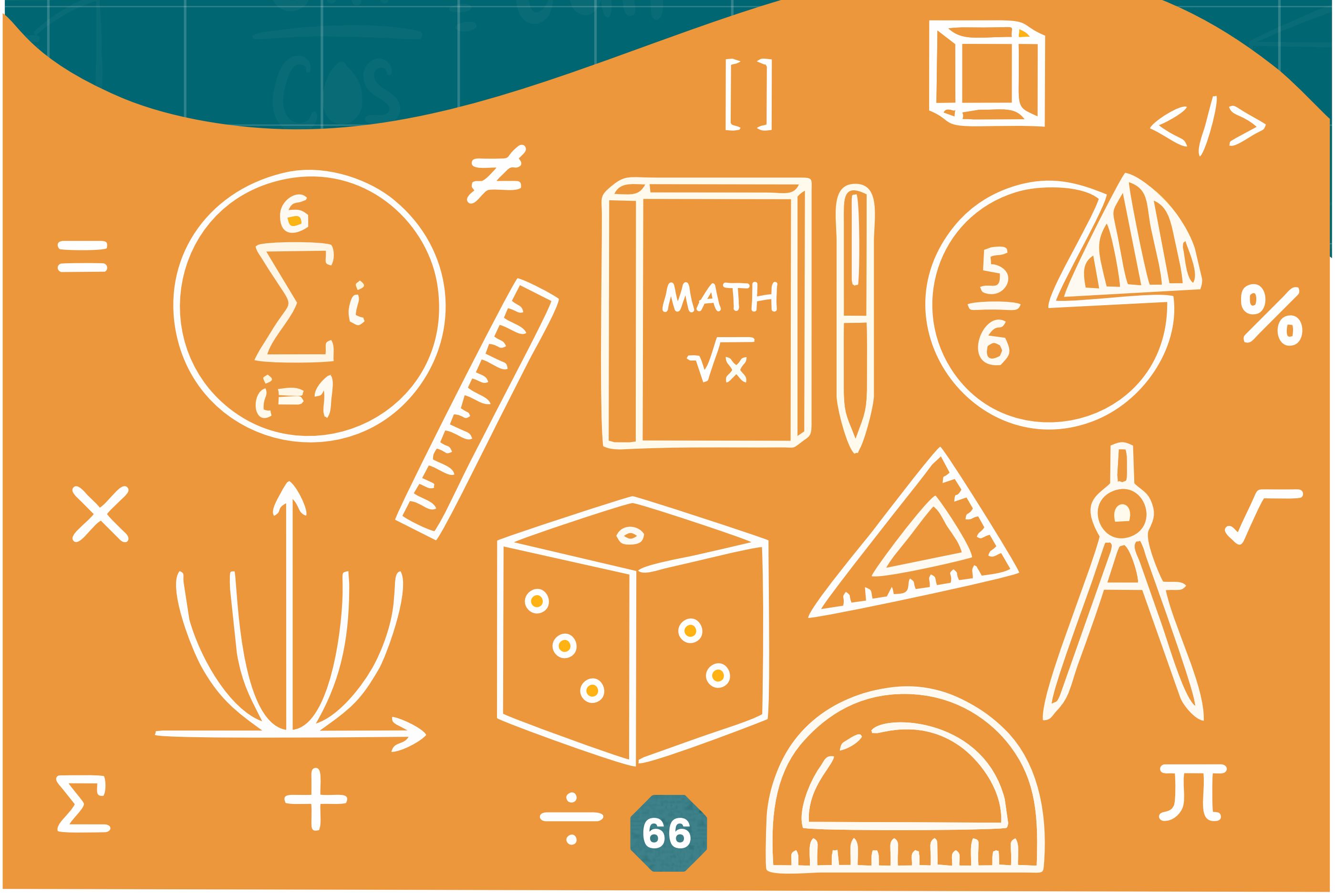
UNIT 7

FINANCIAL ARITHMETIC



Introduction:
In our daily life, we use different facilities provided by the state like police for security, government schools for education, government hospitals for health, etc. Government of the state provides such facilities by an amount which is collected from the citizen of the country and this amount is called **tax.**

TAXES

A fee charged (levied) by the government on a product, income, or activity is called tax. If tax is levied directly on personal or corporate income, then it is called direct tax. If tax is levied on the price of goods or services, then it is called indirect tax.

The purpose of taxation is to finance government expenditure. One of the most important uses of taxes is to finance public goods and services.

7.1.1 Property Tax and General Sales Tax (GST)

There are many types of taxes charged by the government, such as wealth tax, income tax, property tax, general sales tax, etc. Here we should study only the following two types of taxes:
(i) Property Tax
(ii) General Sales Tax (GST)

(i) Property Tax

Property tax is a levy issued by a government on a person's real or personal property. The property is assessed to give it a value and then that value is taxed.
The amount of tax on a given property may change over time based on a re-assessment of the property value.
Property tax payments are due annually in most cases.
It is generally paid at a flat rate of 2% but the tax rates vary, depending on the province. In our province, property tax is levied and collected under the Sindh Urban Property Tax Act 1958.
Following properties are exempted from tax:

Government-owned properties.
Residential building on land not over 120 square yards.
One residential flat of 600 square feet or less on any floor of a building.
Libraries, parks, playgrounds, places of worship, orphanages, cemeteries, crematoriums, etc.
Properties owned by widows, orphaned children, and disabled individuals within some specified limit.
Properties listed as protected heritage under the Sindh Cultural Heritage Preservation Act 1994.

Example 1: Find the property tax on a property of Rs 1,400,000 at the rate of 3%.

Solution:

Worth of the property = Rs 1,400,000

Tax rate = 3%

Property tax = ?

Formula for finding property tax = $\dfrac{(\text{Rate} \times \text{Value of Property})}{100}$

Property tax = 3% of Rs 1,400,000 = $\dfrac{(3 \times 1{,}400{,}000)}{100}$ = Rs 42,000

Thus, the required property tax is Rs 42,000.

7.1.2 Calculation of General Sales Tax (GST)

Since the year 2014, the rate of GST in Pakistan is 17%.

> The GST is calculated as under:
> Total GST = Item Cost × Rate of GST
> So, Rate of GST = $\dfrac{\text{Total GST Tax}}{\text{Item Cost}}$

Example: The cost of a pair of shoes is Rs 2500. The seller charges 17% GST on the cost. Find GST levied.

Solution:

Cost of item = Rs 2,500

Rate of GST = 17%

Value of GST = Item Cost × 0.17
Value of GST = Rs 2,500 × 0.17 = Rs 425
So, the GST charged is Rs 425.

(i) To Determine the Total Cost

In order to determine the total cost and tax inclusive of an item, the GST is added to the item cost.

$$\boxed{\text{So, Total Item Cost} = \text{GST} + \text{Item Cost}}$$

In the above example:
Item Cost = Rs 2,500
Value of GST = Rs 425
Total cost of pair of shoes = Rs 2,500 + Rs 425 = Rs 2,925

$$\boxed{\begin{array}{l}\text{GST} = \text{Total Cost} - \text{Item Cost}\\ \text{Item Cost} = \text{Total Cost} - \text{GST}\end{array}}$$

Let item cost = y
Since Rate of GST = 17% = 0.17
So, GST = 0.17 × y = 0.17y

As we know that:

Item Cost = Total Cost – GST
There for y = Total Cost - 0.17×y

Or Total Cost = y+0.17y = (1 + 0.17)y

Or 1.17y = Total Cost

Or $y = \dfrac{\text{Total cost}}{1.17}$

Hence, $\boxed{\text{Item Cost} = \dfrac{\text{Total Cost}}{1.17}}$, where Rate of GST is 17%

Example 1: A washing machine is sold for Rs 15,912, inclusive of GST. Find the cost of washing machine excluding GST when rate of GST is 17%.

Solution:

$$\text{Item Cost} = \frac{\text{Total Cost}}{1.17}$$

Total cost = Rs 15,912

Rate of GST = 17% = 0.17

Cost of washing machine = $\frac{\text{Rs } (15{,}912)}{1.17}$ = Rs 13,600

Thus, the cost of washing machine (exclusive of tax) is Rs 13,600.

Example 2: Mr. Adam paid Rs 595 as GST at the rate of 17% on the purchase of a shirt. Find the cost of the shirt exclusive of GST.

Solution:

GST = Item Cost × 17%

Here, GST = Rs 595

Cost of shirt = ?

595 = Cost of shirt × 0.17

Cost of shirt = $\frac{\text{Rs } 595}{0.17}$ = Rs 3,500

Thus, the cost of shirt is Rs 3,500.

EXERCISE 7.1

A. Find property tax on the property at the rate mentioned below:

1. Property worth Rs 286,400, rate = 5%.
2. Property worth Rs 1,657,900, rate = 2%.
3. Property worth Rs 814,500, rate = 3%.

B. Find the worth of the property in the following when:

1. Tax paid on the property is Rs 25,500 at the rate of 2%.
2. Tax paid on the property is Rs 35,900 at the rate of 4%.
3. Tax paid on the property is Rs 4,800 at the rate of 5%.

C. Find rate percentage of property tax when:

1. The property tax paid is Rs 30,000 and the worth of the property tax paid is Rs 1,800,000.
2. The property tax paid is Rs 33,000 and the worth of the property is Rs 3,300,000.
3. The property tax paid is Rs 12,000 and the worth of the property is Rs 600,000

EXERCISE 7.2

1. The cost of a parker pen is Rs 875. Find the GST charged on it at the rate of 17%.

2. The cost of a mobile is Rs 4,850. Find the GST at the rate of 17%.

3. The total cost of cosmetic items is Rs 14,976. If the actual cosmetic item cost is Rs 12,800, find the amount of GST.

4. A microwave costs Rs 15,200. Rate of GST is 17% charged on it. Find the amount of GST and total cost of the microwave.

5. Abdullah purchased a three-piece suit of cost Rs 7,800. Find the total cost and amount of GST at the rate of 17%.

6. A refrigerator is sold inclusive of 17% GST for Rs 35,375. Find the actual cost of the refrigerator.

PROFIT AND MARKUP

Profit:

Profit is the amount of gain received from any business, after deducting all cost and expenses from the total sale.

i.e $\boxed{\text{Profit} = \text{Selling Price} - \text{Cost Price}}$

Markup:

In our daily life, we often borrow money from our friends and relatives to buy a thing that we repay them after a certain period. Some banks and retail organization also provide the same services and charge an additional amount called markup.

To find the rate of profit and markup per annum:

Profit Rate:

Profit rate is a profitability ratio that shows the relationship between gross profit and total net sales revenue.

$$\text{Profit Rate} = \frac{(\text{Gross Profit})}{\text{Net Sale}} \times 100$$

Gross profit = Net Sale − Cost of goods

Selling Price = Cost Price + Cost Price × Profit Rate

$$\boxed{SP = CP\,(1 + \text{Profit Rate})}$$

and
$$\boxed{\text{Cost Price} = \frac{\text{Selling Price}}{(1 + \text{Profit Rate})}}$$

Example. Find the profit rate when gross profit is Rs 400 and net sale is Rs 1600.

Solution:

$$\text{Profit Rate} = \frac{\text{Gross Profit}}{\text{Net Sale}} \times 100$$

$$= \frac{\overset{1}{\cancel{400}} \times \overset{25}{\cancel{100}}}{\cancel{1600}_{\cancel{4}_{1}}} = 25\%$$

Thus, Profit Rate = 25%.

Markup of per annum:

The markup as a percentage of the actual amount paid for the things. This is called the markup rate and paid amount itself is called the principal.

Example: Find the markup on a principal amount of Rs 50,000 for 2 years at the rate of 12%.

Solution:

$$\text{Markup} = \frac{P \times R \times T}{100} = \frac{50000 \times 12 \times 2}{100}$$

$$= \text{Rs } 12{,}000$$

Example 1: A person purchased a certain item from wholesaler for Rs 400. Then he sells it to a consumer for Rs 700. What is his profit? Also find his profit rate.

Solution:

We know that Profit = Selling Price – Cost Price

Hence, Profit = Rs 700 – Rs 400 = Rs 300

Now, $\text{Profit Rate} = \dfrac{\text{Profit} \times 100}{\text{Cost Price}}$

Or $\text{Profit Rate} = \dfrac{300}{400} \times 100 = 75\%$

Thus, the profit rate is 75% and his profit is Rs 300

Example 2: A computer software retailers uses a profit rate of 45%. Find the selling price of an electronic computer game that costs the retailer Rs 1250.

Solution:

We know that

Selling Price = Cost Price (1 + Profit Rate)

Here, Selling Price = ?

Cost Price = Rs 1250

Profit Rate =45% =0.45

S.P = C.P (1 + Profit rate)

So, Selling Price = Rs 1250 (1 + 0.45) = Rs (1250 × 1.45) = Rs 1812.50

Thus the Selling Price of the computer game is Rs 1812.50

Example 3: A shoes store uses a 40% profit rate on cost price. Find the cost of a pair of shoes that is sold for Rs 2,630.

Solution:

We know that,

Selling Price = Cost Price (1 + Profit Rate)

Selling Price = Rs 2630

Profit Rate = 40% = 0.4

2630 = CP (1 + 0.4)

$CP = Rs \dfrac{2630}{1.4} = Rs\ 1{,}879$

Hence, the cost of a pair of shoes is Rs 1879.

Remember: The profit is also calculated as a percentage of the original cost price.

$$\text{Profit Rate} = \frac{\text{Profit}}{\text{Cost Price}} \times 100$$

Example 4: A Tile company has Rs 1,500,000 of sales in the previous month. As well as sale returns (means the sale returned back by the consumer due to some reasons) of Rs 45,000, a cost of goods sold for Rs 650,000 and administrative expenses of Rs 470,000. Calculate the net profit percentage of the company.

Solution:

Here is the calculation of profit percentage and related tables.

Net Sale = Rs 1,500,000 (Sales) – Rs 45,000 (Sale returned)
= Rs 1,455,000

Gross Profit = Rs 1,455,000 (Net Sale) – [Rs 650,000 (Cost of goods)
– Rs 470,000 (Administrative expenses)]
= Rs 335,000

Formula: $\text{Profit Rate} = \dfrac{\text{Gross Profit}}{\text{Net Sale}} \times 100$

$$\text{Profit Rate} = \frac{335{,}000}{1{,}455{,}000} \times 100 = 23\%$$

The profit percentage of Tile Company is 23%.

Activity 1. Find the missing quantities by using formula.

S. No.	Net Profit	S.P	C.P	Formula
(i)		Rs 1750	Rs 1600	Profit = S.P - C.P
(ii)	Rs 600		Rs 2000	S.P = C.P + Profit
(iii)	Rs 950	Rs 10,450		S.P - Profit = C.P

Activity 2. Find the missing quantities by formula.

Table 2: Profit Rate Calculation

S.P	C.P	Net Profit	Profit Rate	Formula
Rs 1000	Rs 800			Profit Rate = (Profit / S.P) × 100
	Rs 1000	Rs 100	10%	S.P = C.P (1 + Profit rate)
Rs 2000	Rs 1600	Rs 400		Profit Rate = (Profit / C.P) × 100
Rs 1200	Rs 1000		20%	
Rs 2750	Rs 2500	Rs 250		

Activity 3. Find the missing quantities by using the formula.

Table 3: Markup Analysis

S. No.	Markup	Principal	Time Period	Markup Rate	Formula
(i)		Rs 500	2 years	12%	MP = P × R × T / 100
(ii)	Rs 205		1 year	8%	P = MP× 100/R × T
(iii)	Rs 528	Rs 1,650	10 years		MPR = MP × 100/P×T
(iv)	Rs 350	Rs 3,500		2.5%	T = MP × 100/ R × T
(v)		Rs 100,000	3 years	1.25%	MP = P×R×T/100
(vi)	Rs 1,050		5 years	4.5%	P = MP × 100/R× T

EXERCISE 7.3

Q1. A retailer pays to its whole-saler Rs,1860 for a Helmet and then he sells it to consumer for Rs 2,650. What is the markup rate and his profit.

Q2. A Shopkeeper sells a microwave oven for Rs 6700. Find th markup rate and his profit if the cost price of microwave oven is Rs 6250.

Q3. A retailer uses markup rate of 65% on the computer accessories. Find the selling price if the cost of a certain accessory is Rs 1520.

Q4. The cost price of a certain novel is Rs 1,100. Find selling price of the novel at 66% markup rate.

Q5. A mobile phone company uses a markup rate of 55% on the cost of each mobile phones that cost Rs 2,260. Find the selling price of the mobile phone.

Q6. A mobile phone company uses a 45% markup rate. Find the cost of a unit that sells for Rs 4785.

Q7. The markup rate on the selling price of a calculator that is Rs 1,230 is 50%. Find the cost price of the calculator.

Q8. A stationer uses a markup of 60% on each item that he sells. If an item was sold for Rs 144, Find the cost price of the item.

Real life problems involving markup

Suppose 'P' is the principal amount, "T" is the time period and 'R' is the markup rate. Then the amount of markup will be:

$$\text{Markup} = \frac{R \times P \times T}{100}$$

In the same way we can find:

$$\text{Principal Amount} = \frac{\text{Mark up} \times 100}{R \times T}$$

$$\text{Rate of markup} = \frac{\text{Mark up} \times 100}{P \times T}$$

$$\text{and Time period} = \frac{\text{Mark up} \times 100}{P \times R}$$

Example 1. A person borrowed a loan of Rs 280,000 from a bank at the markup rate of 12% per annum. Find the value of markup, if the time period is 2 years.

Solution:

Principal amount = Rs 280,000Markup rate = 12% annum

Time period (T) = 2years Value of Markup = ?

Formula:

$$\text{Value of the markup} = \frac{R \times P \times T}{100} = \frac{12 \times 280{,}000 \times 3}{100} = \text{Rs}100{,}800$$

Thus the value of the markup is Rs 100,800.

Example 2. Aslam borrowed a loan of Rs 450,000 from a retail organiser for 292 days at the rate of 10% per annum. Find the value of markup.

Solution:

Principal (P) = Rs 450,000, Markup rate (R) = 10% per annum, Markup = ?

$$\text{Time period} = 292 \text{ days} = \frac{292}{365}\text{years} = \frac{4}{5}\text{year}$$

$$\text{By using the formula: Markup} = \frac{R \times P \times T}{100}$$

$$= \frac{10 \times 450{,}000 \times \frac{4}{5}}{100} \quad \frac{\overset{2}{\cancel{10}} \times 450{,}000 \times 4}{\underset{1}{\cancel{100}} \times \underset{1}{\cancel{5}}} = \text{Rs } 36{,}000$$

Thus the value of markup is Rs 36,000

Example 3. On a bank loan, calculate the markup on a principal amount of Rs 87,600 for 20 months at the rate of 12.5% per annum.
Solution:
Principal amount (P) = Rs 87600
Markup rate (R) = 12.5%

Time period (T) = 20 months = $\frac{20}{12}$ years = $\frac{5}{3}$ years

By using the formula: Markup = $\frac{R \times P \times T}{100}$

$$\text{Markup} = \frac{12.5 \times 5 \times \overset{292}{\cancel{87600}}}{\underset{1}{\cancel{100}} \times \underset{1}{\cancel{3}}} = \text{Rs } 18250$$

Hence the markup is Rs 18,250

EXERCISE 7.4

Q1. Qasim bought a car worth Rs 450,800 from a bank loan on instalments at the markup rate of 12% per annum. Find the sale price of the car, if the time period is 3 years.

Q2. Najma bought a golden ornament set for Rs 175,500, on installments at the markup rate of 10% per annum. Find her sale price of the set if the time period is 9 months.

Q3. Shazia sold his mobile phone for Rs 16,900 and got 15% markup rate per annum. Find her markup where the period is 219 days.

Q4. If 15% markup on a colour TV set is 8,500 for 1½ years. Find the cost price of the colour TV set.

Q5. Find the missing quantities by using formula:

S. No.	Markup	Principal	Time Period	Markup Rate
(i)	Rs	Rs 800,000	1 year	12%
(ii)	Rs 10,000	Rs 500,000		10%
(iii)	Rs 6,600		2 year	11%
(iv)		Rs 200,000	1 ½ year	8%
(v)	Rs 800	Rs 4,000	2 ½ year	
(vi)	Rs 45,000		73 days	10%

ZAKAT AND USHR

(i) Zakat

Zakat is one of the five pillars of Islam. It has been mentioned along with Salat in Quran.
The zakat is paid after a complete lunar year at the rate of 2.5%.

$$\boxed{\text{Zakat} = \text{Annual saving} \times 2.5\%}$$

(ii) Ushr

Ushr is paid on compulsory basis at the rate of 5% (or 10%) of the produce by every land owner, grantee, allottee, lessee, lease-holder or land holder. An individual farmer is exempted from ushr if he is a 'mustahiq' or his produce is less than 'Five Wasaqs or 948 kg (approximately) of wheat or its equivalent in value'. If the farm or garden is watered naturally by rain or spring water or river or stream water, tenth part or 10% is obligatory, and if farm or garden is irrigated by artificial means like well, tube well, canals etc. twentieth part or 5% is obligatory.

(iii) Distribution of Zakat and Ushr

ZAKAT IS PAID:

- to the poor and needy.
- to those who are in bondage and debt and for wayfarer.
- to those who are employed to administer the funds.

Example 1: After a whole year a business man has saved Rs 80,000 in cash. He has a cash in Bank of Rs 600,000. Find the Zakat due on him. Solution:

(i) Saving in cash = Rs 80,000

(ii) Cash in bank = Rs 600,000

Total saving of the businessman = 80,000 + 600,000

= Rs 680,000

Amount of Zakat due on him = Rs 680,000 × 2.5%

$= 680{,}000 \times \frac{2.5}{100} =$ Rs 17,000

The Zakat due on the businessman is Rs 17,000.

Example 2: A household woman has gold ornaments of value of Rs 500,000. She also saved an amount of Rs 25,000 from daily life expenses. Find the Zakat due upon her.

Solution:

Total amount = Rs 500,000 + Rs 25,000 = Rs 525,000

So, Zakat = Rs 525,000 × 2.5%

$=$ Rs $525{,}000 \times \frac{2.5}{100} =$ Rs 13,125

Thus, she has to pay Rs 13,125 as Zakat.

Example 3: Farah paid an amount of Rs 9,000 as Zakat. Find the saving on which she paid the Zakat.

Solution:

We know that: Amount of Zakat = Annual saving × 2.5%

9,000 = Annual saving × 0.025

Annual saving $= \frac{9{,}000}{0.025} =$ Rs 360,000

Thus she had an annual saving of Rs 360,000.

EXERCISE 7.5

Q1. A person holds an amount of Rs 500,000 for the whole year. Is he sahib-e-nisab? If yes, find the amount of Zakat he has to pay.

Q2. A salaried person possess Rs 200,000 and 3 tolas of pure gold. Find the Zakat to be paid by him. Rate of gold is Rs 47,400 per tola.

Q3. A lady doctor possesses ornaments worth Rs 500,000 and saved Rs 150,000 in the whole year. Find the amount of Zakat due upon her.

Q4. A businessman paid Rs 20,500 as Zakat. Find his saving for the whole year.

Q5. A farmer pays Rs 10,000 as Ushr. Find the value of the crop he produced. Rate of Ushr is 5%.

Q6. A farmer produces 15,000 kg of sugarcane. Find his amount of Ushr at the rate of 5%. The rate of sugarcane is Rs 5.25 per kg.

Q7. Jumman exports oranges of his garden for Rs 5,000,000. Find the amount of Ushr at the rate of 5%.

Q8. Kamil produces 1,500 kg of wheat on one part of his land and 1,200 kg of rice on the other part of his land. Find the amount of Ushr at the rate of 5%. The rate of wheat is Rs 30 per kg and rate of rice is Rs 65 per kg.